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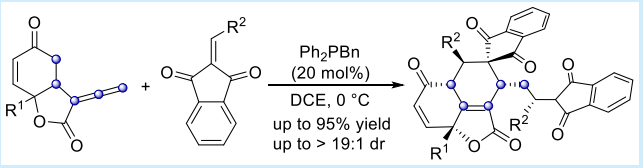
Phosphine-Catalyzed Cascade Michael Addition/[4+2] Cycloaddition Reaction of Allenoates and 2-Arylidene-1,3-indanediones

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ABSTRACT: The phosphine-catalyzed cascade Michael addition/[4+2] cycloaddition reaction of tetrahydrobenzofuranone-derived allenoates and 2-arylidene-1,3-indanediones has been reported, affording spirocyclic 1,3-indanedione derivatives in moderate to high yields with moderate to good diastereoselectivities. A scaled-up reaction worked well under mild conditions, and a plausible mechanism is proposed.

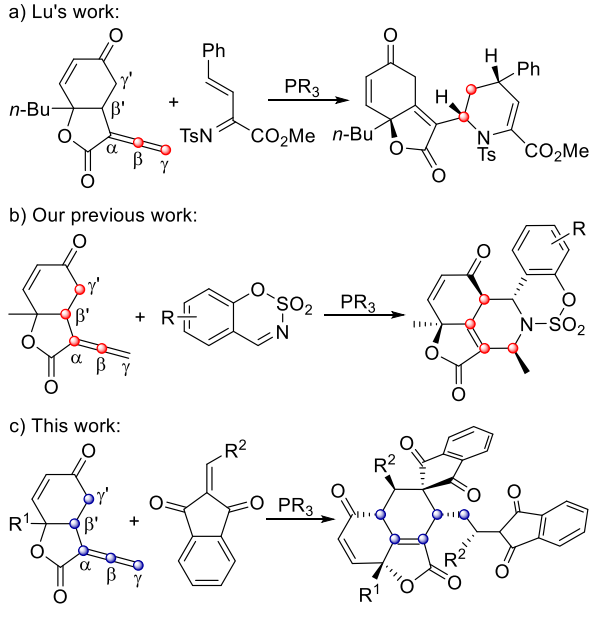


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Nucleophilic phosphine catalysis remains an active area of organocatalysis.[1] Particularly interesting phosphine-catalyzed cycloaddition reactions have attracted more attention and have been developed as an efficient access to a variety of synthetically useful carbo- and heterocycles. In general, nucleophilic attack of phosphine on an electron-deficient phosphine acceptor generates zwitterionic intermediates, which were then in situ trapped with electron-poor unsaturated substrates leading to synthetic approaches to carbo- and heterocycles. Obviously, the phosphine acceptor plays a key role in these annulation reactions. Therefore, development of nucleophilic phosphine catalysis is highly dependent on the evolution of the phosphine acceptor. A handful of phosphine acceptors, including electron-deficient alkenes, alkynes, and allenes, have been exploited for phosphine catalysis.[1] Among these phosphine acceptors, allenes are very versatile precursors with multiple reactivities, which have been successfully used in many types of catalytic cycloaddition reactions. However, the synthetic potential of classical allenes in phosphine-catalyzed cycloaddition reactions has nearly been exhausted; therefore, introduction of an additional functional group into allenoates, giving them a new function, will offer certain opportunities for development of novel reactions.

Recently, the Lu group and our group reported a type of allenoate having an electron-withdrawing group at the γ'-carbon of the allenoates (Scheme 1).[2,3] With this kind of tetrahydrobenzofuranone-derived allenoate as the phosphine acceptor, the Lu group achieved phosphine-catalyzed [4+2] annulation involving α,β-unsaturated imine to give a chiral tetrahydropyridine derivative in high yield with excellent enantioselectivity, in which the allenoate worked as a C2 synthon (Scheme 1a).[2] Our group developed phosphine-catalyzed [4+2] cycloaddition of this tetrahydrobenzofuranone-derived allenoate and sulfamate-derived cyclic imines.[3] As opposed to the previous cycloaddition reaction of allenoates, the γ'-carbon of the allenoate was involved in this reaction (Scheme 1b).[3] In most phosphine-catalyzed cycloaddition reactions reported so far involving allenes, allenes were used as one-membered,[4] two-membered,[5] three-membered,[6] and four-membered[7] reactants, leading to the construction of cyclic systems in which the allene unit is

Scheme 1. Previous Works and Our Strategy



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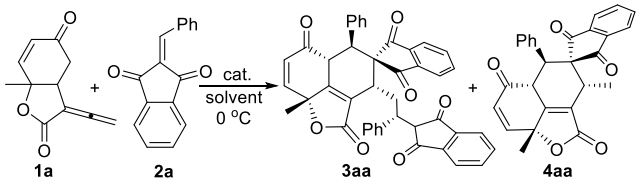
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partly or completely fused in the resulting ring. In comparison, the use of the allenes involving more than four carbon atoms (for instance, five) in cycloaddition reactions has not been described. With our previous work as a basis,[2,8] herein we report a phosphine-catalyzed cascade Michael addition/[4+2] cycloaddition of the tetrahydrobenzofuranone-derived allenoates and 2-arylidene-1,3-indanediones, giving valuable spirocyclic 1,3-indanedione derivatives[9] (Scheme 1c).

In our initial attempt, the model reaction of allenoate **1a** and 2-benzylidene-1*H*-indene-1,3(2*H*)-dione **2a** was investigated in CH_2Cl_2 at 0 °C. A 1.2:1 molar ratio of allenoate (**1a**, 0.12 mmol) to alkene (**2a**, 0.1 mmol), which was often used in phosphine, was utilized. Under catalysis with 20 mol % Ph_3P, an unexpected tandem reaction product **3aa** was isolated in 25% yield with a trace amount of [4+2] cycloadduct **4aa**. Therefore, as shown in Table 1, a 1:2 molar ratio of allenoate (**1a**, 0.1 mmol) to alkene (**2a**, 0.2 mmol) was used in the next screening. With this molar ratio, Ph_3P promoted the cascade reaction to give the product **3aa** in 62% yield with good diastereoselectivity as well as a [4+2] cycloadduct **4aa** in 22% yield (entry 1). The stronger nucleophilic phosphine Ph_2PMe led to the product **3aa** in 72% yield, and only a trace amount of **4aa** was observed (entry 2). Surprisingly, the strong nucleophilic phosphines $PhPMe_2$ and *n*-Bu_3P did not show catalytic reactivity, and no product was observed (entries 3 and 4, respectively). On the basis of these results, several diarylphosphines and triarylphosphines such as (4-$MeC_6H_4)_3P$, (4-$FC_6H_4)_3P$, and Ph_2PBn were examined. When (4-$MeC_6H_4)_3P$ was used, the product **3aa** was obtained in 60% yield with a 7:1 dr and the byproduct **4aa** was isolated in 22% yield (entry 5). By contrast, (4-$FC_6H_4)_3P$ showed similar catalytic reactivity, giving a 63% yield of **3aa** with a 6:1 dr and a 24% yield of **4aa** (entry 6). To our delight, Ph_2PBn displayed both good catalytic activity and chemical selectivity, delivering an 84% yield of **3aa** with a 13:1 dr and only a trace of **4aa** (entry 7). Further catalyst screening indicated that tertiary amines such as 4-dimethylaminopyridine (DMAP), 1,4-diazabicyclo[2.2.2]octane (DABCO), and 1,8-diazabicyclo-[5,4,0]undec-7-ene (DBU) did not catalyze the reaction (entries 8−10, respectively). Investigation of solvents indicated that 1,2-dichloroethane (DCE) is most suitable for this cascade reaction, affording the product **3aa** in an 88% yield with a 15:1 dr (entry 11). Toluene (entry 12) gave a result similar to those of DCE and DCM did (entries 7 and 11, respectively). Polar solvent CH_3CN led to an acceptable yield of only 45% of **3aa** with a 10:1 dr (entry 13). Increasing or decreasing the reaction temperature did not improve the results (entries 14 and 15), and 0 °C was a better choice in terms of yield and reaction time. Decreasing the catalyst loading to 10% still resulted in the product **3aa** in an 85% yield with a 13:1 dr, albeit requiring a longer reaction time (entry 16). Very interestingly, even using a 1:1 molar ratio of **1a** to **2a**, only a trace amount of the [4+2] cycloadduct **4aa** was observed, although the yield of tandem reaction product **3aa** was reduced to 35% (entry 17). On the basis of the results presented above, the optimal reaction condition was determined to be one with Ph_2PBn (20 mol %) as the catalyst in 1 mL of DCE at 0 °C. The relative configuration of the annulation product has been unambiguously confirmed by X-ray crystallography of products **3aa** and **4aa**.[11]

Table 1. Optimization of the Reaction Conditions[a]



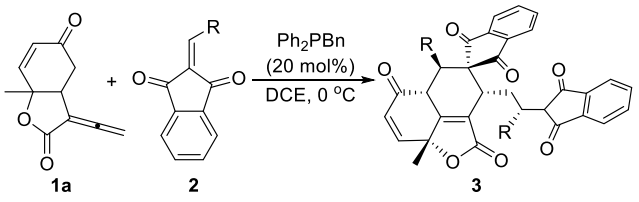
entry	catalyst	solvent	3aa/4aa yield (%)[b]	*t* (h)	dr[c]
1	Ph_3P	CH_2Cl_2	62/22	1	8:1
2	Ph_2PMe	CH_2Cl_2	72/trace	24	7:1
3	$PhPMe_2$	CH_2Cl_2	NR	24	–
4	*n*-Bu_3P	CH_2Cl_2	NR	24	–
5	(4-$MeC_6H_4)_3P$	CH_2Cl_2	60/22	1	7:1
6	(4-$FC_6H_4)_3P$	CH_2Cl_2	63/24	1	6:1
7	Ph_2PBn	CH_2Cl_2	84/trace	1	13:1
8	DMAP	CH_2Cl_2	NR	24	–
9	DABCO	CH_2Cl_2	NR	24	–
10	DBU	CH_2Cl_2	NR	24	–
11	Ph_2PBn	DCE	88/trace	1	15:1
12	Ph_2PBn	toluene	82/trace	1	11:1
13	Ph_2PBn	CH_3CN	45/trace	72	10:1
14[d]	Ph_2PBn	DCE	82/trace	0.5	10:1
15[e]	Ph_2PBn	DCE	88/trace	24	14:1
16[f]	Ph_2PBn	DCE	85/trace	24	13:1
17[g]	Ph_2PBn	DCE	35/trace	1	13:1

[a]Unless otherwise stated, reactions of **1a** (0.10 mmol) and **2a** (0.2 mmol) were carried out at 0 °C in the presence of a catalyst (0.02 mmol) in 1 mL of the solvent. [b]Isolated yield. [c]Determined by ^1H NMR analysis. [d]The reaction was performed at 25 °C. [e]The reaction was performed at −10 °C. [f]With 10 mol % catalyst. [g]With 0.1 mmol of **2a**.

With the optimal condition in hand, we next explored the scope of 2-arylidene-1,3-indanediones (Table 2). The results showed that both electron-deficient and -rich substituents on the benzene ring are suitable for this cascade reaction (entries 1−16). Most halogen-substituted substrates worked well to give the desired products in good yields with moderate to good diastereoselectivities (8:1 to 15:1 dr, ≤95% yield, entries 1−6). A special dichloro-substituted 2-arylidene-1,3-indanedione **2j** showed moderate reactivity, giving the desired product **3aj** in a 59% yield with a 15:1 dr (entry 9). Electron-deficient olefins bearing strong electron-withdrawing groups on the benzene such as 4-nitro-substituted olefin **2h** and 4-trifluoromethyl-substituted olefin **2i** displayed high reactivities, affording the desired products **3ah** and **3ai** in 83% and 72% yields, respectively, with good diastereoselectivities (both >19:1 dr, entries 7 and 8, respectively). A majority of 2-arylidene-1,3-indanediones with electron-donating groups also showed good reactivities. 2-Methyl-, 3-methyl-, 4-methyl-, and 3-methoxy-substituted 2-arylidene-1,3-indanediones underwent the cascade reaction efficiently with 78%, 83%, 78%, and 81% yields, respectively, with moderate diastereoselectivity (entries 10−12 and 14, respectively). In particular, 4-ethyl-substituted 2-arylidene-1,3-indanedione **2n** afforded the product **3an** in a 90% yield and a relatively better diastereoselectivity (entry 13). Disappointingly, 4-methoxy-substituted and 2,5-dimethyl-substituted olefins **2p** and **2q** showed low reactivity; the target products **3ap** and **3aq** were produced in acceptable yields of 48% and 56%, respectively (entries 15 and 16, respectively). 2-(1-Naphthylvinyl)-1,3-indanedione **2r** also underwent this reaction, and the desired product **3ar** was obtained in a 69% yield with a 10:1 dr (entry 17). Unfortunately, styryl-substituted 2-arylidene-1,3-indanedione did not undergo this

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Table 2. Substrate Scope of 2-Arylidene-1,3-Indanedione[a]

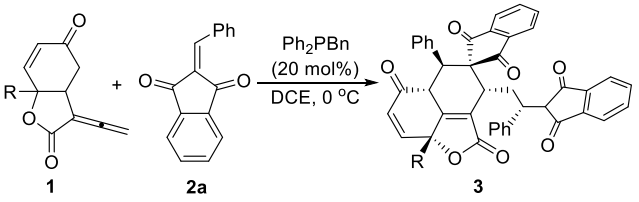


entry	R in **2**	**3**	*t* (h)	yield (%)[b]	dr[c]
1	2-FC_6H_4 (**2b**)	**3ab**	1.5	95	8:1
2	3-FC_6H_4 (**2c**)	**3ac**	1	88	15:1
3	4-FC_6H_4 (**2d**)	**3ad**	1	80	15:1
4	4-ClC_6H_4 (**2e**)	**3ae**	1	84	14:1
5	4-BrC_6H_4 (**2f**)	**3af**	1	87	13:1
6	4-IC_6H_4 (**2g**)	**3ag**	2	84	12:1
7	4-$CF_3C_6H_4$ (**2h**)	**3ah**	1	83	>19:1
8	4-$NO_2C_6H_4$ (**2i**)	**3ai**	3	72	>19:1
9	2,4-$Cl_2C_6H_3$ (**2j**)	**3aj**	2	59	15:1
10	2-MeC_6H_4 (**2k**)	**3ak**	1	78	8:1
11	3-MeC_6H_4 (**2l**)	**3al**	1	83	11:1
12	4-MeC_6H_4 (**2m**)	**3am**	1	78	12:1
13	4-EtC_6H_4 (**2n**)	**3an**	2	90	15:1
14	3-$OMeC_6H_4$ (**2o**)	**3ao**	2	81	9:1
15	4-$OMeC_6H_4$ (**2p**)	**3ap**	1.5	48	10:1
16	2,5-$Me_2C_6H_3$ (**2q**)	**3aq**	2	56	12:1
17	1-naphthyl (**2r**)	**3ar**	11	69	10:1
18	styryl (**2s**)	**3as**	24	trace	–

[a]All reactions were carried out with **1a** (0.1 mmol), **2** (0.2 mmol), and Ph_2PBn (20 mol %) in 1 mL of DCE at 0 °C. [b]Isolated yield. Only a trace of product **4** was detected for all cases. [c]Determined by ^1H NMR analysis.

reaction (entry 18). As shown in Table 3, several different alkyl-substituted allenoates **1b**–**1d** were examined, affording

Table 3. Substrate Scope of Allenoates[a]



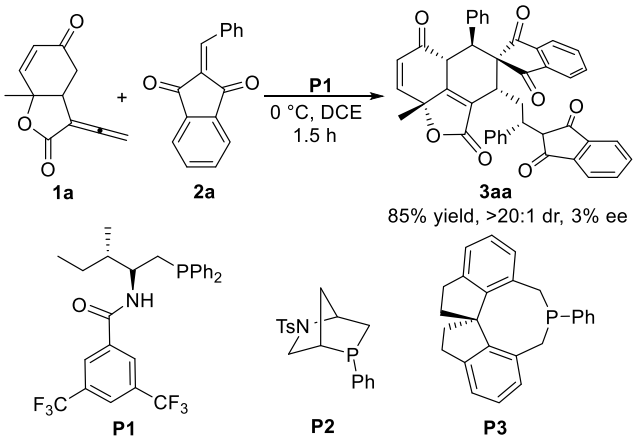
entry	R in **1**	**3**	yield (%)[b]	dr[c]
1	Et (**1b**)	**3ba**	84	12:1
2	*n*-Pr (**1c**)	**3ca**	59	6:1
3	*i*-Pr (**1d**)	**3da**	89	10:1

[a]All reactions were carried out with **1** (0.1 mmol), **2a** (0.2 mmol), and Ph_2PBn (20 mol %) in 1 mL of DCE at 0 °C. [b]Isolated yield. A trace of product **4** was detected for all reactions. [c]Determined by ^1H NMR analysis.

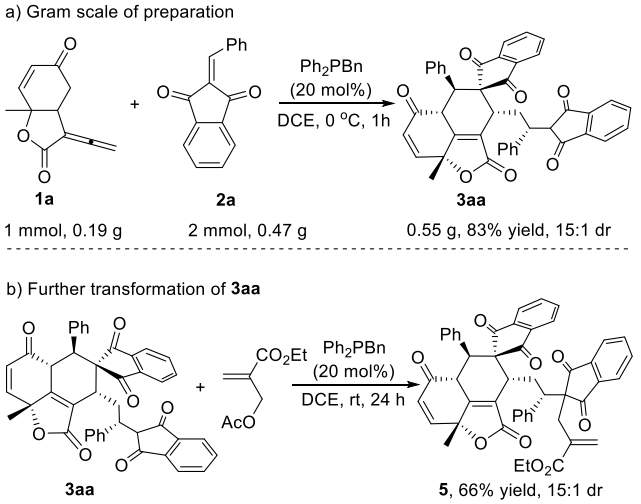
the corresponding product **3ba**–**3da**, respectively, in moderate to good yields with good diastereoselectivities.

As shown in Scheme 2, this reaction with an asymmetric variant was not successful. Using the optimal chiral bifunctinal phosphine **P1**,[10a] an only 3% ee was obtained. Other chiral phosphines such as **P2**[6e] and **P3**[10b] led to a trace amount of the product. To demonstrate the synthetic potential of the current cascade reaction, a scale-up reaction was performed (Scheme 3a). The tetrahydrobenzofuranone-derived allenoate **1a** (1 mmol, 0.19 g) was reacted with 2-arylidene-1,3-

Scheme 2. Attempt with an Asymmetric Variant



Scheme 3. Scale-Up Reaction and Further Transformation of the Product

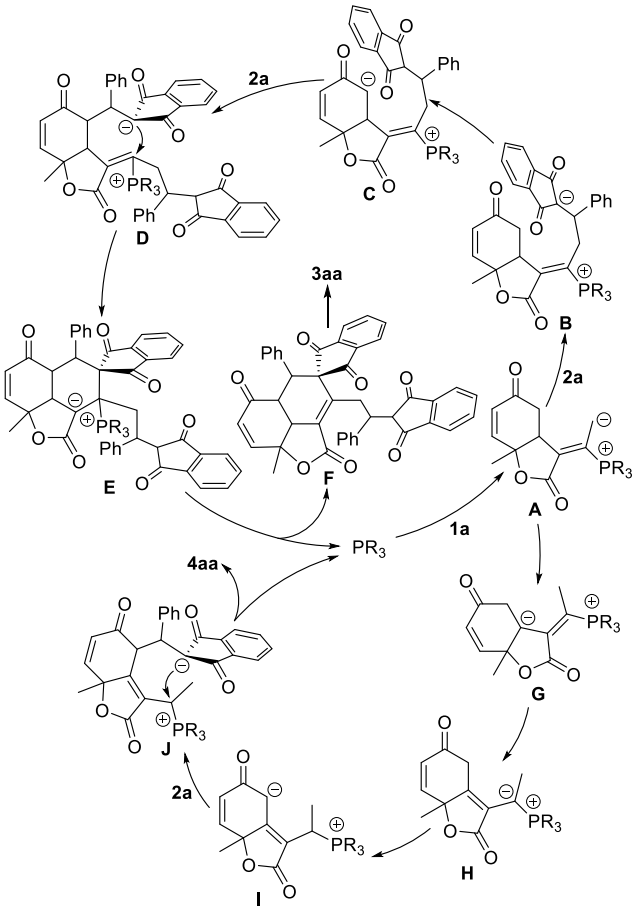


indanedione **2a** (2 mmol, 0.47 g) under the optimal condition, giving the product **3aa** in an 83% yield with a 15:1 dr. Compared with the reaction on a 0.1 mmol scale, no obvious loss of yield or diastereoselectivity was observed. In the presence of ethyl 2-(acetoxymethyl)acrylate and Ph_2PBn, allylation of the product **3aa** occurred in DCE at room temperature to give adduct **5a** in a 66% yield with a 15:1 dr.

On the basis of the general law of phosphine-catalyzed reactions,[1,3] a plausible mechanism was proposed (Scheme 4). At first, nucleophilic attack of a phosphine catalyst on tetrahydrobenzofuranone-derived allenoate **1a** leads to formation of zwitterionic intermediate **A**, which attacks 2-arylidene-1,3-indanedione **2a** to give intermediate **B**. Then, intermediate **B** undergoes a proton migration to afford intermediate **C**, which attacks another 2-arylidene-1,3-indanedione **2a** to produce intermediate **D**. Then an intramolecular cycloaddition occurs to generate intermediate **E**. Subsequent formation of a double bond leads to regeneration of the phosphine catalyst with simultaneous generation of intermediate **F**, which isomerizes to tandem reaction product **3aa**. In another catalytic cycle, intermediate **A** performs a consecutive proton transfer or isomerization to give intermediate **I**, which attacks electron-deficient alkene **2a** to produce intermediate **J**. Consequent intramolecular nucleophilic substitution results in [4+2] cycloaddition product **4aa**.

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Scheme 4. A Plausible Reaction Mechanism



In conclusion, a phosphine-catalyzed cascade Michael addition/[4+2] cycloaddition reaction of allenoates and 2-arylidene-1,3-indanedione has been achieved under mild conditions, affording diverse multisubstituted spirocyclic 1,3-indanedione derivatives in good yields with moderate to good diastereoselectivities. It is worth mentioning that five carbon atoms were involved in this tandem reaction, offering allenoates as phosphine acceptors with more features.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.0c00637.

Experimental procedures, characterization data, HPLC analysis data, and NMR spectra (PDF)

Accession Codes

CCDC 1898871 and 1921150 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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